VoltRelay Energy

Network Performance, Service Reliability & Rider Retention Analysis

Data Analytics Hackathon — Final Report

2.97M	~94.7%	12.39%	87.79%
Cleaned Swap Events	Completion Rate	Peak Monthly Failure Rate	Kyron Retirement Rate

1. Executive Summary

This report analyzes VoltRelay Energy's battery-swapping network across six Indian cities — Bengaluru, Delhi NCR, Hyderabad, Pune, Mumbai, and Jaipur — using eight operational datasets covering swap events, station status, riders, batteries, support tickets, and city-level context. After data cleaning and exclusion of two test stations, the analysis is built on **2,970,790** cleaned swap events.

The network's failure rate is normally stable, sitting in the **4.2%–4.4%** range during typical months. However, the data shows a sharp reliability deterioration between April and June 2024, peaking at **12.39%** in May 2024 — roughly three times the baseline rate — before returning to normal levels by July 2024. This is the single most significant operational finding in the dataset.

Failures are not evenly distributed. **Jaipur (6.40%)** and **Delhi NCR (5.98%)** show the highest city-level failure rates, and many of the highest-failure individual stations are **Gen1/Launch** infrastructure concentrated in these two cities. Gen1 stations as a class show a higher failure rate (**5.91%**), slower charging (**91.8 minutes**), and lower charged-battery availability (**8.28** average) compared to Gen2 and Gen3 stations.

On the equipment side, battery supplier analysis reveals a strong asset-quality signal: while Amptek and Cellora batteries show **0% retirement**, **Kyron** batteries show an **87.79% retirement rate** and a current average SOH of only **63.75%**, despite starting at a comparable ~99% initial SOH. Drilling into manufacturing lots shows this issue is concentrated in three specific lots — **KY-2407, KY-2408, and KY-2409** — each with retirement rates above 97%, while later Kyron lots (KY-2410 through KY-2418) show 0% retirement in the available data.

Rider retention, by contrast, is not a significant concern in this dataset: observed 30-day retention among eligible new riders is **99.60%**, with only **63** non-returning riders out of **15,784** eligible. This sample is too small to draw conclusions about churn drivers, and it reframes the business priority away from retention and toward operational reliability and battery/equipment quality, which show much stronger and more actionable signals.

The report's core recommendations are to (1) prioritize station-level investigation in Jaipur and Delhi NCR, (2) audit and prioritize upgrade planning for high-volume Gen1 stations, (3) investigate the Kyron KY-2407/2408/2409 manufacturing lots with the supplier, (4) improve charging turnaround and ready-to-swap battery availability, and (5) establish ongoing monitoring of failure rate, battery-lot retirement, and station-level performance rather than relying on periodic analysis alone.

2. Business Problem

VoltRelay Energy operates a battery-swapping network serving 2-wheeler (2W) and 3-wheeler (3W) gig-economy and logistics riders across six Indian cities. While completed swaps and revenue have grown, the business has simultaneously experienced rising service failures, concerns about new-rider retention, and pressure on unit economics. Management needed an evidence-based view of what is actually driving these issues before committing to infrastructure or supplier-related investment decisions.

This analysis was structured around six questions: (1) How has network performance evolved over time? (2) Where and why do service failures occur, and how do they affect customer experience? (3) Which stations and cities underperform, and why? (4) Is battery and charger equipment quality contributing to failures? (5) Is new-rider retention actually a problem? (6) What operational actions follow from the evidence?

3. Data Overview

The analysis draws on eight datasets covering the full operational surface of the network: swap transactions, station infrastructure and hourly status, rider records, battery assets, support tickets, and city-level daily context.

Dataset	Rows	Cols	Contents
swap_events.csv	3,877,013	22	Individual swap attempts (completed and failed)
station_hourly_status.csv	1,487,712	14	Hourly station-level operating status
riders.csv	20,000	12	Rider profile and signup attributes
batteries.csv	6,500	13	Battery asset master data (SOH, supplier, lot)
support_tickets.csv	44,000	12	Customer support ticket records
stations.csv	152	22	Station master data (city, generation, capacity)
city_daily_context.csv	3,282	12	City-level daily operating context
fleet_partners.csv	12	12	Fleet partner master data

All row/column counts above reflect the raw, as-provided dataset sizes prior to cleaning.

4. Data Cleaning & Preparation

Before analysis, the datasets were put through a structured cleaning process covering duplicate checks, missing-value assessment, date-range validation, categorical-value validation, and cross-dataset relationship/integrity checks. Relationship integrity checks between datasets returned **0 invalid references**, indicating strong referential consistency across the tables.

Additional corrections applied during cleaning included text standardization (e.g., inconsistent city-name variants), correction of invalid SOC/SOH values, correction of negative distance values, and a firmware-related timestamp correction applied to **135,659** events.

Two non-operational test stations — **STN-TST-01** and **STN-TST-02** — were identified and excluded from all operational performance analysis. After cleaning and test-station exclusion, the working swap-events dataset (**swap_events_clean**) contains **2,970,790 rows × 22 columns**.

The cleaned dataset breaks down into the following event types:

Event Type	Count
swap_completed	2,814,959
failed_no_charged_battery	88,160
abandoned_queue	45,991
cancelled_by_rider	12,779
failed_system_error	8,901

5. Network Performance Analysis

Across the cleaned dataset of **2,970,790** swap events, the overall completion rate is approximately **94.7%**. Under stable conditions, the monthly failure rate consistently sits in the **4.2%–4.4%** range. This baseline, however, was disrupted by a sharp deterioration between April and June 2024.

Month	Failure Rate
Jan 2024	4.41%
Feb 2024	4.42%
Mar 2024	4.39%
Apr 2024	6.88%
May 2024	12.39%
Jun 2024	10.28%
Jul 2024	4.33%
Aug 2024	4.27%
Sep 2024	4.38%
Oct 2024	4.37%
Nov 2024	4.24%
Dec 2024	4.39%
Apr 2025 (partial month)	6.92%

Key Finding — Mid-2024 Reliability Deterioration

The failure rate rose from a stable ~4.4% baseline to a peak of **12.39%** in May 2024 (nearly 3x baseline), before recovering to normal levels by July 2024. April 2025 is a partial month and is reported for reference only — it is not directly comparable to full calendar months.

The May 2024 spike was driven primarily by two failure categories: **failed_no_charged_battery** (**11,192** events in May 2024) and **abandoned_queue** (**5,452** events in May 2024). Together with cancelled_by_rider and failed_system_error, these four categories account for all recorded failure types in the cleaned dataset. The concentration of the spike in charged-battery shortages and queue abandonment points toward a supply-side (battery readiness) and queueing issue during this window, rather than a broad-based system failure.

6. Geographic & Station Performance

Failure rates vary meaningfully by city. Jaipur and Delhi NCR show the highest observed failure rates, while Bengaluru and Mumbai show the lowest.

City	Attempts	Failure Rate
Jaipur	379,826	6.40%
Delhi NCR	595,483	5.98%
Hyderabad	549,989	5.53%
Pune	434,234	4.70%
Mumbai	415,193	4.45%
Bengaluru	596,065	4.45%

Station-level analysis of the highest-failure stations shows that many of them are **Gen1/Launch** stations, and this pattern is particularly pronounced in Jaipur and Delhi NCR — the same two cities with the highest city-level failure rates. This suggests failures are geographically concentrated and linked to where older infrastructure is deployed, rather than spread evenly across the network.

7. Charger Generation & Infrastructure Analysis

Stations were grouped by charger generation (Gen1, Gen2, Gen3) and compared on failure rate, charged-battery availability, and charging time.

Generation	Stations	Attempts	Failure Rate	Avg Charged 2W	Avg Charge Time
Gen1	51	1,502,163	5.91%	8.28	91.8 min
Gen2	62	1,309,037	4.58%	9.04	64.2 min
Gen3	34	159,590	4.42%	9.91	42.1 min

Gen1 stations show a higher failure rate, lower average charged-battery availability, and substantially longer charging times than Gen2 and Gen3. This is an **observed association** between charger generation and reliability outcomes, not proof that older charger hardware directly causes failures — other correlated factors (station age, location, usage load) may also contribute. It is also worth noting that Gen3 has considerably lower attempt volume (**159,590**) than Gen1 (**1,502,163**) and Gen2 (**1,309,037**), so generation comparisons should account for this difference in sample size before being used to justify large capital decisions.

8. Battery Supplier & Manufacturing-Lot Analysis

Battery health was compared across the network's three suppliers: Amptek, Cellora, and Kyron. All three suppliers started at a comparable initial state-of-health (SOH) of approximately 99%.

Supplier	Batteries	Avg Initial SOH	Avg Current SOH	Retirement Rate
Amptek	1,307	99.19%	81.01%	0%
Cellora	3,555	99.18%	80.93%	0%
Kyron	1,638	99.21%	63.75%	87.79%

Key Finding — Kyron Asset-Quality Signal

Despite a comparable starting SOH (~99%), Kyron batteries show a current average SOH of only **63.75%** and a retirement rate of **87.79%**, versus 0% for Amptek and Cellora. This is a strong observed association between the Kyron supplier and battery deterioration/retirement, and it warrants direct investigation with the supplier — it is not, on its own, proof of a manufacturing defect or causal fault.

Manufacturing-Lot Detail

Breaking Kyron batteries down by manufacturing lot shows the deterioration is not uniform across all Kyron batteries — it is concentrated in three specific, large lots.

Manufacturing Lot	Batteries	Avg Current SOH	Retirement Rate
KY-2407	480	60.99%	99.17%
KY-2408	500	61.05%	97.80%
KY-2409	481	60.92%	98.34%
KY-2410 – KY-2418	177	~86.5%	0%

Lots KY-2407, KY-2408, and KY-2409 each show retirement rates above 97%, while the later lots (KY-2410 through KY-2418) show 0% retirement in the available battery data. This is a **lot-level quality signal** — it suggests the issue is tied to specific production batches rather than to the Kyron brand as a whole.

Swap-level records for these three lots tell a complementary story: at the time swaps were recorded, batteries from KY-2407, KY-2408, and KY-2409 showed average SOH-in values of **86.44%**, **86.48%**, and **86.51%** respectively (across 177,129, 184,331, and 177,153 swap events), with an average distance since previous swap of roughly 54–55 km. These swap-level SOH figures are noticeably higher than the ~61% current SOH shown in the battery master table for the same lots. This is not a contradiction — the two figures represent different points in the battery lifecycle: swap-level SOH reflects battery condition at the moment of historical swap transactions, while current SOH in the battery master table reflects a later (and in this case, degraded) lifecycle state closer to retirement.

9. Rider Retention Analysis

A 30-day retention analysis was run on new riders eligible for the observation window.

Metric	Value
Eligible new riders	15,784
Returned within 30 days	15,721
Observed 30-day retention	99.60%
Non-returning riders	63

Observed 30-day retention is **99.60%**, which is very high. However, this result should be read carefully: the non-returning group is only **63 riders** out of 15,784 eligible, which is too small a sample to support strong conclusions about what drives churn. The descriptive breakdown below is presented for completeness, not as evidence of causal churn drivers.

Descriptive Profile of the 63 Non-Returning Riders

Vehicle Class & Plan Type

Vehicle	Count
2W	49
3W	14

Plan Type	Count
Partner billed	36
Pay as you go	20
Prepaid pack	7

Signup Channel & Home City

Channel	Count
Partner onboarding	27
Field agent	16
App store	13
Referral	7

City	Count
Bengaluru	18
Hyderabad	13
Pune	11
Delhi NCR	10
Jaipur	5
Mumbai	3

Note: home-city figures reflect the naming variants present in the source data (e.g., BLR / Bangalore / New Delhi were treated as the same city). These counts describe who did not return; they should not be interpreted as evidence that vehicle class, plan type, signup channel, or city causes churn, given the very small sample size.

Because observed retention is 99.60%, the operational reliability and battery-quality findings in this report are considerably stronger and more actionable analytical signals than rider retention. Retention should be monitored going forward but is not, on this data, a primary driver of current business pressure.

10. Visualization Findings

Three primary visualizations were produced from the cleaned data, supported by an executive KPI summary.

Visualization	What It Shows
Monthly Network Failure Rate	Shows the stable ~4.2–4.4% baseline and the sharp April–June 2024 spike, peaking at 12.39% in May 2024.
Network Failure Rate by City	Shows Jaipur and Delhi NCR as the highest-failure cities, with Bengaluru and Mumbai the lowest.
Failure Rate by Charger Generation	Shows Gen1 stations with the highest observed failure rate, and Gen3 with the lowest.

An executive KPI dashboard was also produced, summarizing cleaned swap events (2.97M), overall completion rate (~94.7%), peak monthly failure rate (12.39%), and Kyron battery retirement rate (87.79%) in a single view for stakeholder reference.

11. Five Key Insights

1. Mid-2024 Service Reliability Deterioration

Network failure rate rose from a ~4.4% baseline to 12.39% in May 2024, driven mainly by failed_no_charged_battery and abandoned_queue events, before recovering by July 2024.

2. Geographic Concentration of Failures

Failures are not evenly distributed: Jaipur (6.40%) and Delhi NCR (5.98%) show materially higher failure rates than the network average, with the highest-failure stations concentrated in these cities.

3. Gen1 Infrastructure Reliability Gap

Gen1 stations show a higher failure rate (5.91%), slower charging (91.8 min), and lower charged-battery availability than Gen2 and Gen3, an observed association pointing to aging infrastructure as a contributing factor.

4. Kyron Battery Asset-Quality Signal

Kyron batteries show an 87.79% retirement rate and 63.75% current SOH versus 0% retirement for Amptek and Cellora, despite comparable starting SOH — a strong signal warranting supplier investigation.

5. Concentration of Kyron Retirement in Specific Manufacturing Lots

The Kyron issue is concentrated in lots KY-2407, KY-2408, and KY-2409 (97%+ retirement each), while later lots KY-2410–KY-2418 show 0% retirement, pointing to a lot-level rather than brand-wide quality issue.

12. Recommendations

The following recommendations follow directly from the evidence above. They describe priority areas for management attention and are not guaranteed to fully resolve the underlying issues; each should be validated operationally before major investment.

1. Investigate high-failure stations in Jaipur and Delhi NCR

Prioritize root-cause investigation and operational improvements at the highest-failure stations in these two cities, where failure rates and Gen1 concentration are both elevated.

2. Audit and prioritize upgrades for high-volume Gen1 stations

Conduct technical audits of high-volume Gen1 stations and build upgrade business cases, given their observed association with higher failure rates and slower charging.

3. Investigate Kyron manufacturing lots KY-2407, KY-2408, and KY-2409

Engage the Kyron supplier on a lot-level quality investigation for these three batches, which account for the overwhelming majority of observed battery retirement.

4. Improve battery charging turnaround and ready-to-swap availability

Address charging turnaround time and charged-battery inventory levels, particularly at Gen1 stations, given their link to failed_no_charged_battery events.

5. Build ongoing operational monitoring

Establish continuous monitoring of station failure rates, battery-lot retirement, SOH deterioration, queue wait times, and battery availability, rather than relying on periodic point-in-time analysis.

13. Limitations

- The dataset is synthetic; findings describe patterns within this dataset and should be validated against real operational data before major decisions are made.
- This is an observational analysis. Associations described in this report (charger generation, supplier, geography) do not establish causation.
- April 2025 is a partial month and is not directly comparable to full calendar months in the network performance trend.
- The rider non-return sample (63 riders) is very small; churn-driver conclusions cannot be reliably supported from this dataset.
- Battery current SOH (master data) and swap-level SOH represent different points in the battery lifecycle and should not be treated as the same measurement.
- Supplier and charger-generation differences should be validated with additional operational data before major capital investment decisions are made.
- The cleaned swap-event file's coverage ends earlier than the June 2025 period referenced in the original problem statement, so this analysis should not be presented as a complete view of network performance through June 2025.

14. Conclusion

This analysis of VoltRelay Energy's battery-swapping network identifies two dominant, well-evidenced issues: a mid-2024 reliability shock concentrated in charged-battery shortages and queue abandonment, and a set of infrastructure and equipment quality patterns — older Gen1 stations and specific Kyron battery manufacturing lots — that are associated with elevated failure rates and battery deterioration. Both are geographically concentrated, with Jaipur and Delhi NCR standing out consistently across the station, city, and generation views. Rider retention, while often a default area of concern for platform businesses, is not supported as a current problem by this data: observed

30-day retention is 99.60%, with too few non-returning riders to draw further conclusions. The evidence points management toward operational and equipment-quality interventions — station-level investigation, Gen1 infrastructure upgrades, and supplier/lot-level battery investigation — as the highest-value next steps, supported by ongoing monitoring rather than one-time analysis.

Management Takeaway

The network's core problem is operational and equipment-driven, not rider churn: a mid-2024 reliability shock and a Gen1/Kyron-battery quality pattern — concentrated in Jaipur and Delhi NCR — are the strongest, most actionable findings in this data. Management should prioritize station-level investigation in these two cities, a Gen1 upgrade business case, and a supplier investigation into Kyron lots KY-2407–KY-2409, backed by ongoing monitoring rather than periodic review. Rider retention (99.60% observed) does not warrant near-term intervention on this evidence.